47-851: Modern Convex Optimization Lecture 0

Modern Convex Optimization

Lecturer: Karan Singh

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47-851: Modern Convex Optimization Lecture 1

Introduction and Complexity of Local Minimization

Lecturer: Karan Singh

This is a fast-paced course that develops the basic toolkit of convex optimization. We will see what makes convexity an attractive lens to examine optimization problems, and develop efficient algorithms. A good part of the course will also be spent on (conceptual) applications in applied mathematics and engineering. In roughly five weeks, we will visit:

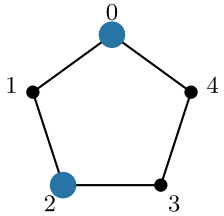
1. Convex sets and functions
2. Duality: Conic, Lagrangian, Fenchel
3. Structured conic optimization: theory, models, applications
4. Large-scale optimization: ellipsoid, first-order methods, acceleration, lower bounds

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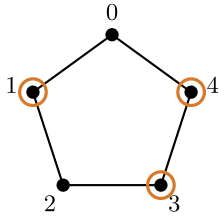
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1. A Motivating Example

Consider a language or a script with n alphabets. Some letter pairs look alike. If one were to communicate using alphabets alone, one must avoid the use of at least one alphabet in all such letter pairs. How many letters are usable?



A stable set: $\{0, 2\}$



Its complement is a vertex cover

Consider a graph $G = (V, E)$ where V is the set of letters and $(u, v) \in E$ if and only if u and v look alike; this notion is symmetric. We are looking for the largest *stable set*.

$$\alpha(G) = \max_{S \subseteq V} \{|S| : S \text{ contains at most one endpoint of every edge}\}.$$

Other related graph-theoretic notions:

- Clique is a subset of vertices that are all pairwise connected.
- Vertex Cover is a subset of vertices such that for every edge, at least one endpoint is in it.

Exercise. Prove that S is an independent set on G iff S is a clique on the edge-complement of G.

Exercise. Prove that S is an independent set on G iff $V - S$ is a vertex cover on G.

Another interesting fact is that the maximum number of maximum (in fact, maximal too) independent sets is $3^{\frac{n}{3}} \ll 2^n$. Can you think of graphs for which this is attained?

A question of Claude Shannon (1956) concerns "worst-case information theory": How much more efficient are words than letters? If one were to communicate with k-letter words, $|V| = n^k$, and 2 words $(\alpha_1 ... \alpha_k)$, $(\beta_1, ... \beta_k)$ would be ambiguous if and only if they are not the same, yet for all positions $1 \leq \ell \leq k$, α_ℓ and β_ℓ are ambiguous (or the same). Drawing edges between such words, we get G^k. Graph theorists call it the "strong graph product," denoted by $\boxtimes$, applied k times.

Exercise. How do the adjacency matrices of G and G^k relate? Think of Kronecker products.

We expect $\alpha\big(G^k\big)$ to scale exponentially in k; after all the vertex set does. So to normalize this effect, let us define

$$\Theta(G) = \lim_{k \to \infty} \sqrt[k]{\alpha(G^k)}.$$

Exercise. Prove that $\alpha\big(G^k\big) \geq (\alpha(G))^k$.

Hint: Using unambiguous letters alone one can construct unambiguous words.

But in fact, $\Theta(G)$ can be quite a bit greater than $\alpha(G)$, indicating words are more communication-efficient than letters. Here was the concrete question Shannon posed.

C_n is the cycle graph with n vertices. Clearly $\alpha(C_5) = 2$. How large is $\Theta(C_5)$?

This was resolved by Laszlo Lovasz, in 1979, 23 years later in a result that is one of many origins of Semi-Definite Programming; this we will study, and you'll prove the result in one of the assignments!

Theorem 1 (Lovasz '79) Using SDPs, we have

$$\Theta(C_5) = \sqrt{5} > 2.$$

An unresolved question even in the LLM age is: what is $\Theta(C_7)$?

2. Preliminaries and Conventions

The following are notions of *similarity* and *length* respectively.

Definition 2 $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$, where V is a vector space, is an inner product if
1. $\langle x, x \rangle \geq 0$ for all $x \in V$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
2. $\langle x, y \rangle = \langle y, x \rangle$ for all $x, y \in V$.
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ for all $x, y \in V$ and $\alpha \in \mathbb{R}$.
4. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ for all $x, y, z \in V$.

Definition 3 $\|\cdot\| : V \to \mathbb{R}$, for vector space V, is a norm if
1. $\|x\| \geq 0$ for all $x \in V$ and $\|x\| = 0$ if and only if $x = 0$.
2. $\|\alpha x\| = |\alpha| \, \|x\|$ for all $x \in V$ and $\alpha \in \mathbb{R}$.
3. $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$.

If the second part of condition (1) fails in either definition, we call these semi inner products and semi norms respectively. Under such definitions, non-zero vectors can have zero *length*.

Lemma 4 (Cauchy-Schwarz) For any (semi) inner product, for all $x, y \in V$

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

Proof. Consider that $\|sx - ty\|^2 \geq 0$ for all $x, y \in V$. We will fix s, t later.

$$s^2 \langle x, x \rangle + t^2 \langle y, y \rangle - 2st \langle x, y \rangle \geq 0$$

$$\Rightarrow \frac{s^2 \langle x, x \rangle + t^2 \langle y, y \rangle}{2} \geq st \langle x, y \rangle.$$

Substitute $s = \frac{1}{\sqrt{\langle x, x \rangle}}$, $t = \frac{1}{\sqrt{\langle y, y \rangle}}$, then we are done unless $\langle x, x \rangle$ or $\langle y, y \rangle$ is zero. □

Exercise. Complete the above proof for cases when $\langle x, x \rangle$ or $\langle y, y \rangle$ is zero.

Hint: Think about what happens if we choose arbitrarily large s or t in these cases.

Proposition 5 $f(x) = \sqrt{\langle x, x \rangle}$ is a norm.

Proof. (1) & (2) are straightforward. For all $x, y \in V$, we have

$$\begin{aligned} f(x + y)^2 &= \langle x, x \rangle + \langle y, y \rangle + 2 \langle x, y \rangle \\ &\leq \langle x, x \rangle + \langle y, y \rangle + 2 f(x) f(y) \quad \text{(C.S.)} \\ &\leq (f(x) + f(y))^2. \end{aligned}$$

□

2.1. When do minimizers exist?

For practical optimization problems, a desirable property is that there is indeed a best decision, no matter how difficult or easy it is to find.

Definition 6 A set S is closed if for any (convergent) sequence contained within it, the corresponding limit also lies in it.

A set S is bounded if $\exists C \in \mathbb{R}$ such that $\forall x, y \in S$, $\|x - y\| \leq C$.

Compact = bounded and closed.

A generic optimization problem looks like

$$\mathcal{O}(f, K) \equiv \min_{x \in K} f(x).$$

Definition 7 $\mathcal{O}(f, K)$ is feasible if K is not empty.

$\mathcal{O}(f, K)$ is bounded below if $\exists C \in \mathbb{R}$, such that $\forall x \in K, f(x) \geq C$.

$\mathcal{O}(f, K)$ is solvable if $\exists x^* \in K$ such that $\forall x \in K, f(x) \geq f(x^*)$.

For linear programming:

$$\text{feasible \& bounded below} \Rightarrow \text{solvable.}$$

But **false** for Convex programs: $\min_{x \in \mathbb{R}} e^{-x}$.

This is the cause of a lot of pain when stating any non-LP duality result.

Theorem 8 If K is compact & non-empty and f is continuous, then a minimizer exists.

This theorem is nice, but we would also like something that applies to unconstrained problems.

Theorem 9 If K is closed & non-empty, f is continuous with all its sub-level sets bounded, then a minimizer exists.

Definition 10 α-sublevel set of f on K for any $\alpha \in \mathbb{R}$ is $S_\alpha(f) = \{x \in K : f(x) \leq \alpha\}$.

3. Convexity of Sets and Functions

Consider V to be any vector space and K to be any convex set so that adding elements makes sense.

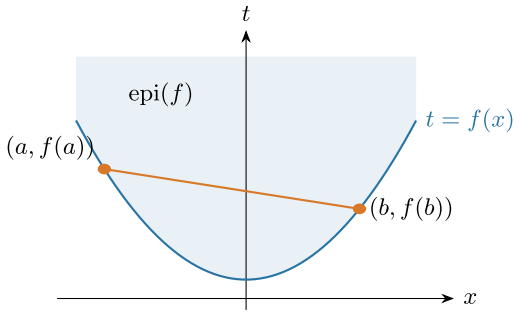
Definition 11 (Convex Set) $S \subseteq V$ is convex if $\forall x, y \in S$, $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda) y \in S$.

Definition 12 (Convex Function) $f : K \to \mathbb{R}$ is convex if $\forall x, y \in K$, $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y).$$

The set $\{\lambda x + (1-\lambda)y : \lambda \in [0,1]\}$ is the line segment connecting x and y. So convex sets are precisely those that contain all line segments for which corresponding endpoints are within the set. And convex functions are precisely those for which any linear interpolation overapproximates the function value. How are these related?

Definition 13 (Epigraph) $\text{epi}(f) = \{(t,x) : t \geq f(x)\}$.



Theorem 14 f is a convex function if and only if $\text{epi}(f)$ is a convex set.

Proof. $\Rightarrow$ Take $(t,x),(s,y) \in \text{epi}(f)$. Then

$$\lambda(t,x) + (1-\lambda)(s,y) = (\lambda t + (1-\lambda)s, \lambda x + (1-\lambda)y) \in \text{epi}(f)$$

since

$$\begin{aligned} \lambda t + (1-\lambda)s &\geq \lambda f(x) + (1-\lambda)f(y) \quad \text{by membership} \\ &\geq f(\lambda x + (1-\lambda)y). \end{aligned}$$

$\Leftarrow$ Take $x, y \in K$, $\lambda \in [0,1]$. Note that $(f(x),x),(f(y),y) \in \text{epi}(f)$. Hence

$$(\lambda f(x) + (1-\lambda)f(y), \lambda x + (1-\lambda)y) \in \text{epi}(f).$$

Thus $\lambda f(x) + (1-\lambda)f(y) \geq f(\lambda x + (1-\lambda)y)$. □

One attraction of convexity is that it is a robust notion, that is, it is preserved under many (useful) operations. We will describe a few operations for convex sets below. You should be able to translate these to convex functions via the above link.

Exercise. Prove that Cartesian products, Minkowski sums, linear transformations and intersections of convex sets are convex.

If you have not seen this before, $A + B = \{a + b : a \in A, b \in B\}$ is a Minkowski sum.

Exercise. Construct a closed set $S \subseteq \mathbb{R}^n$, and a matrix $A \in \mathbb{R}^{m \times n}$ such that the linear transformation $AS = \{Ax : x \in S\}$ is not closed.

Hint: Linear transformations can be used to do projections onto subspaces.

Exercise. Prove $S_\alpha(f)$ is a convex set $\forall \alpha \in \mathbb{R}$ if f is a convex function. Is the converse true?

4. Local and Global Minimizers

Definition 15 $\overline{x}$ is a local minimum for f on K if $\exists \varepsilon > 0$ such that

$$f(\overline{x}) \leq f(x) \quad \forall x \in K \cap B(\overline{x}, \varepsilon).$$

$\overline{x}$ is a strict local minimum for f on K if $\exists \varepsilon > 0$ such that

$$f(\overline{x}) < f(x) \quad \forall x \in K \cap B(\overline{x}, \varepsilon) - \{\overline{x}\}.$$

$\overline{x}$ is a global minimum for f on K if $\forall x \in K$

$$f(\overline{x}) \leq f(x).$$

$\overline{x}$ is a strict global minimum for f in K if

$$\forall x \in K - \{\overline{x}\}, \quad f(\overline{x}) < f(x).$$

Here and elsewhere, $B(\overline{x}, \varepsilon) = \{x : \|x - \overline{x}\| \leq \varepsilon\}$. Think of $\|\cdot\|$ as the Euclidean norm, but for finite-dimensional vector spaces, all norms are equivalent (sandwich each other up to constants), so it does not matter. These definitions apply to any (for example, non-convex) function. Also note that global minima are local minima.

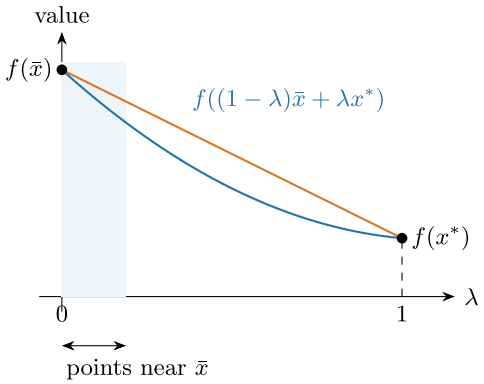
Exercise. Give a non-constant function over $K = \mathbb{R}$ all of whose local minima are global *maxima*.

Proposition 16 For a convex function f, any local minimum is also a global minimum.

Proof. Take a local minimum $\overline{x} \in K$. If $\overline{x}$ is not a global minimum, then $\exists x^* \in K$ such that $f(x^*) < f(\overline{x})$. Note x^* need not be a global minimum; a priori it is not clear if the latter exists. Now set $x = (1 - \lambda)\overline{x} + \lambda x^* = \overline{x} + \lambda(x^* - \overline{x}) \in K$; we will ensure $\lambda \in (0, 1]$. We have

$$f(x) \leq (1 - \lambda) f(\overline{x}) + \lambda f(x^*) < f(\overline{x}),$$

yet x can be made arbitrarily close to $\overline{x}$. For example, if we want the distance to be at most ε, we can set $\lambda = \min\left\{1, \frac{\varepsilon}{\|\overline{x} - x^*\|}\right\}$. □



You might expect that finding a local minimum is somewhat easy: given that it imposes only local qualifications, maybe some greedy procedure produces these. If so, for convex functions, the above

proposition translates them to global minima. This is a key attraction of convexity: doing well enough locally ensures global performance. But this intuition is misleading, as we show next.

5. Computational Hardness of Local Minimization

Theorem 17 Given $\overline{x} \in \mathbb{R}^d$, checking if $\overline{x}$ is a local minimum for a degree 4 polynomial is NP-hard.

Theorem 18 Given $\overline{x} \in \Delta$, where $\Delta = \left\{x \in \mathbb{R}^n : x \geq 0, \sum_{i=1}^n x_i \leq 1\right\}$, checking if $\overline{x} \in \Delta$ is a local (constrained) minimum for a degree 2 polynomial is NP-hard.

We will prove a weaker result.

Theorem 19 (STRICT-4-HARD) Checking if $x \in \mathbb{R}^n$ is a local strict minimum for a degree 4 polynomial is NP-hard.

Note that these results concern certification of a given point. It can still be that other local minima are easy to find; this does happen for a few problem classes. But also finding something that is guaranteed to exist (via compactness, say) cannot be NP-hard, which concerns decision problems, not search. These problems are better translated to PPAD-hardness.

5.1. Global problems can be difficult

We will build towards a proof of the last theorem. The easy bit is to believe that global minimization is hard; so let us start there.

Definition 20 **3SAT** is NP-complete and asks if $\exists$ a boolean assignment of $x, y, z...$ such that

$$(x \vee \overline{y} \vee z) \wedge (\overline{x} \vee y \vee \overline{z}) \wedge ...$$

is satisfied. Three literals in each clause are joined by ORs with negations allowed. Clauses themselves are joined by ANDs.

1-in-3SAT is also NP-complete. It asks if there is a boolean assignment that makes exactly one of the literals in each OR clause true.

***d*-POLYPOS** asks if given a degree-d multivariate polynomial (d is a fixed number, so no representational issues here), is it true that $\forall x \in \mathbb{R}^n, P(x) > 0$.

Claim 21 6-POLYPOS is NP-hard.

Proof. We will embed 3SAT in a degree 6-poly.

$$P(x) = \sum_{i=1}^{n} (x_i(1-x_i))^2$$
$$+\underbrace{((x_1 + (1-x_2) + x_3 - 1)(x_1 + (1-x_2) + x_3 - 2)(x_1 + (1-x_2) + x_3 - 3))^2}_{x \vee \overline{y} \vee z} + \ldots$$

There is a degree 6 term for each OR clause.

$P(x) = 0$ if and only if $x_i \in \{0, 1\}$ and for each clause, at least one (signed) literal is true, e.g.,

$$x_1 + (1 - x_2) + x_3 \in \{1, 2, 3\}$$

which is the same as 3SAT being satisfied. □

Claim 22 4-POLYPOS is NP-hard.

Proof. Same proof, except we embed 1-in-3SAT. This time we only need $(x_1 + (1 - x_2) + x_3 - 1)^2$. □

We are butchering complexity-theoretic reductions a bit. We are using Cook/algorithmic reductions, whereas Karp/logical reductions are more standard. But Cook-NP-hard is plenty hard too.

5.2. Embedding global hardness locally

For any degree-d polynomial P, let $P_h(x, y) := y^d P\left(\frac{x}{y}\right)$ be its homogeneous counterpart. Note that $P_h(x, 1) = P(x)$. Also note that this transformation is symbolic; so we can plug in $y = 0$.

Observation: For any homogeneous polynomial P_h, $\forall \alpha \geq 0$, $P_h(\alpha x) = \alpha^d P_h(x)$, where d is the degree.

Claim 23 For a homogeneous poly q, $x = 0$ is a strict local min $\Leftrightarrow q(x) > 0$ for all $x \neq 0$.

Proof. $\Leftarrow$ $q(0) = 0$ always. We are done.

$\Rightarrow$ $\exists \varepsilon > 0$ such $\forall y \in B(0, \varepsilon) - \{0\}$, $q(y) > q(0) = 0$. But then consider any $z \in \mathbb{R}^n$, $z \neq 0$.

$$q(z) = q\left(\frac{\|z\|}{\varepsilon} \times \frac{z}{\|z\|}\varepsilon\right) = \left(\frac{\|z\|}{\varepsilon}\right)^d q\left(\frac{z}{\|z\|}\varepsilon\right) > 0$$

because $\frac{z}{\|z\|}\varepsilon$ belongs to $B(0, \varepsilon)$. □

The reason for considering homogenized polynomials is as follows: they display the same quantitative behavior in a neighborhood of the origin as they display elsewhere, up to suitable scaling. This characteristic can be used to hide global hardness in a small near-zero neighborhood; the latter is what local minima care about.

Finally, we know that testing positivity of degree 4 polynomials is NP-hard. If this were also established for homogeneous equivalents, we would be done. Except this doesn't make sense directly because homogeneous polynomials evaluate to zero at zero. To complete this translation, consider the following assertion for even degree polynomials:

$$P(x) > 0 \quad \forall x \quad \Leftrightarrow \quad P_h(x, y) > 0 \quad \forall (x, y) \neq 0.$$

To be clear, we will fail to prove this as stated, but we will establish this for special cases which will nevertheless suffice for us.

Here, $\Leftarrow$ holds always as $P(x) = P_h(x, 1) > 0$.

$\Rightarrow$ is tricky. Let's try the contrapositive.

$$\exists (x, y) \neq 0 \text{ such } P_h(x, y) \leq 0.$$

We wish to conclude $\exists z$, $P(z) \leq 0$. This is indeed fine if $y \neq 0$ incidentally.

$$P\left(z := \frac{x}{y}\right) = \frac{P_h(x, y)}{y^d} \leq 0.$$

But this doesn't work if $y = 0$. In fact, consider the counterexample below.

Example. $P(x_1, x_2) > 0$ everywhere, yet $P_h(1, 0, 0) = 0$, where

$$P(x_1, x_2) = x_1^2 + (1 - x_1 x_2)^2,$$

$$P_h(x_1, x_2, y) = y^2 x_1^2 + \left(y^2 - x_1 x_2\right)^2.$$

The subtle observation is that the bad case never happens for the polynomial constructed for 1-in-3SAT. Indeed

$$P_h(x, y) = \sum_{i=1}^{n} \left(x_i(y - x_i)\right)^2 + \left(y(x_1 + (y - x_2) + x_3) - y^2\right)^2 + ...$$

With $y = 0$, the first term becomes $\sum_{i=1}^{n} x_i^4$. Hence $P_h(x, 0) \leq 0 \Rightarrow x_i = 0$ for all i. But the only problematic case was when $\exists (x, y) \neq 0$ and $y = 0$ where $P_h(x, y) \leq 0$.

So our proof of the hardness of certifying strict local minima is complete.

47-851: Modern Convex Optimization Lecture 2

Convexity and Separation

Lecturer: Karan Singh

So far, we have seen that an attractive property of convex functions is that local minima are also global minima. However, we have also seen that even local minima can be hard to find and certify generally. So we will begin by relaxing the definition of local minima itself, in a way that retains the local-to-global property while being tractable.

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6. First-Order Conditions for Local Minimization

Definition 24 d is a descent direction of f at x if there exists $\overline{\alpha} > 0$ such that $f(x + \alpha d) < f(x)$ for all $\alpha \in (0, \overline{\alpha})$.

Claim 25 If $\langle \nabla f(x), d \rangle < 0$, then d is a descent direction for f at x.

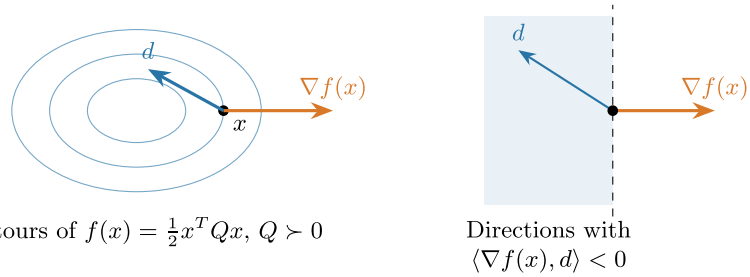
Proof. By Taylor's theorem, we have

$$f(x + \varepsilon d) = f(x) + \langle \nabla f(x), d \rangle \varepsilon + r(\varepsilon, x, d),$$

where $r(x, \varepsilon, d) = o(\varepsilon)$, that is, $\lim_{\varepsilon \to 0^+} r(\varepsilon, x, d)/\varepsilon = 0$. There exists $\overline{\varepsilon} > 0$ such that for all $0 < \varepsilon \leq \overline{\varepsilon}$, $|r(\varepsilon, x, d)/\varepsilon| \leq |\langle \nabla f(x), d \rangle| \, /2$. Then, we get

$$\frac{f(x + \varepsilon d) - f(x)}{\varepsilon} = \langle \nabla f(x), d \rangle + \frac{r(\varepsilon, x, d)}{\varepsilon} \leq \frac{1}{2} \langle \nabla f(x), d \rangle < 0.$$

□



Contours of $f(x) = \frac{1}{2}x^T Qx$, $Q \succ 0$ | Directions with $\langle \nabla f(x), d \rangle < 0$

The converse is false. Consider $f(x) = -x^2$; then at 0 any non-zero direction is a descent direction.

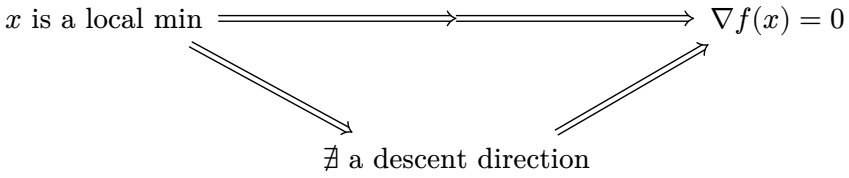
Proposition 26 *x is a local min for $f \Rightarrow \nabla f(x) = 0$.*

Proof. Say $\nabla f(x) \neq 0$; then $\langle \nabla f(x), -\nabla f(x) \rangle < 0$. Hence $-\nabla f(x)/\|\nabla f(x)\|$ is a descent direction. Hence there exists $\overline{\alpha} > 0$ such that for all $0 < \varepsilon < \overline{\alpha}$, there exists $x' \in B(x, \varepsilon)$, namely

$$x' = x - \varepsilon \frac{\nabla f(x)}{\|\nabla f(x)\|},$$

such that $f(x') < f(x)$. Hence x is not a local min. □

To summarize all this, we have:



Exercise. Provide separate counterexamples for each of the three converses.

6.1. The Constrained Case

Now, let's do the constrained case. We have dealt with non-convex functions so far. Coming to the constrained case, you might be tempted to also treat general (non-convex) sets. But, for example, if K is the set of vertices on a hypercube, all feasible points are local minima, regardless of the objective function. To avoid such pathologies, it makes sense to only consider a convex set K.

Proposition 27 *$\overline{x}$ is a local min for f on a convex set $K \Rightarrow \langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0$ for all $x \in K$.*

Proof. Say there exists $x \in K$, $\langle \nabla f(\overline{x}), x - \overline{x} \rangle < 0$. Then consider

$$y = (1 - \lambda)\overline{x} + \lambda x = \overline{x} + \lambda(x - \overline{x}) \in K.$$

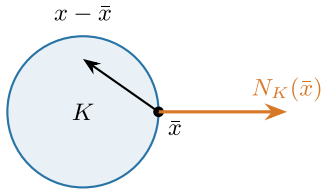
Now, by Taylor's theorem, $f(y) = f(\overline{x}) + \lambda \langle \nabla f(\overline{x}), x - \overline{x} \rangle + r(\lambda, x, \overline{x})$ where $r(\lambda, x, \overline{x}) = o(\lambda)$. Then there exists $\overline{\lambda} > 0$ such that for all $0 < \lambda < \overline{\lambda}$, $|\frac{r(\lambda, x, \overline{x})}{\lambda}| < \frac{1}{2} |\langle \nabla f(\overline{x}), x - \overline{x} \rangle|$. For all such λ's, we have

$$f(y) \leq f(\overline{x}) + \frac{\lambda}{2} \langle \nabla f(\overline{x}), x - \overline{x} \rangle < f(x).$$

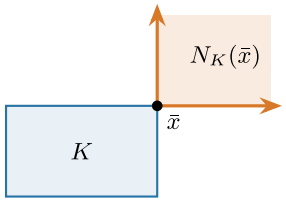
Thus, there does not exist an $\varepsilon > 0$ such that $f(y) \geq f(\overline{x}) \forall y \in B(\overline{x}, \varepsilon)$, because $\lambda(x - \overline{x})$ can be made arbitrarily small, yet $f(\overline{x} + \lambda(x - \overline{x})) < f(\overline{x})$. □

The normal cone of K at x captures all the directions starting at x along which one cannot progress while being restricted to K in a very strong sense, that is, it is not even possible to move along directions which correlate with a member of the cone.

Definition 28 $N_K(\overline{x}) = \{d : \langle d, x - \overline{x} \rangle \leq 0, \forall x \in K\}$ is the normal cone of K at $\overline{x}$.



Smooth boundary: a ray



Corner: a cone of normals

With this definition, we get an alternative characterization:

$$\langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0, \forall x \in K \equiv -\nabla f(\overline{x}) \in N_K(\overline{x}).$$

In words, one would like to go in the direction of $-\nabla f(\overline{x})$ to decrease function value, but all directions that cast a positive shadow on $-\nabla f(\overline{x})$ are unapproachable at $\overline{x}$ by moving in feasible directions.

7. Positive Semidefinite Matrices

Now we have relaxed analogs of local minima which can be easily checked and found. Nevertheless, we have lost something in this relaxation. Can we at least come up with more stringent notions that help us certify local minima in special cases?

To do this, we will consider high-order derivatives, for which the definition below captures the central property of interest.

Definition 29 A symmetric matrix $S \in \mathbb{R}^{n \times n}$ is positive semi-definite if $x^\top S x \geq 0, \forall x \in \mathbb{R}^n$. In this case, we will write $S \succeq 0$. Also $S_+^{n \times n}$ is the space of such matrices.

Similarly, we'll say S is positive definite or $S \succ 0$ if $\forall x \neq 0, x^\top S x > 0$.

Theorem 30 (Spectral Theorem) Any symmetric matrix can be written as

$$S = \sum_{i=1}^{n} \lambda_i u_i u_i^\top$$

where $\lambda_i \in \mathbb{R}$ are eigenvalues, and u_i's are the corresponding eigenvectors (all orthonormal).

Theorem 31 The following statements are all equivalent:
1. $S \succeq 0$.
2. All eigenvalues of S are non-negative.
3. $S = LL^\top$ for some $L \in \mathbb{R}^{n \times k}$ for some k.
4. There exist vectors $v_1, ..., v_n \in \mathbb{R}^n$ such that $S_{ij} = v_i^\top v_j, \forall i, j \in [n]$.
5. Determinants of all principal submatrices are non-negative.

Recall that principal submatrices are those formed by only keeping (or equivalently, deleting) columns and rows corresponding to the same set of indices.

Proof. We will leave out (5), and form a chain from (1) through (4).

(1) $\Rightarrow$ (2) For all x, $x^\top Sx \geq 0$. Take $x = u_i$. Then $0 \leq u_i^\top S u_i = \lambda_i \|u_i\|^2$.

For the next two steps, we reinterpret matrix multiplication two ways. Consider $A \in \mathbb{R}^{n \times k}$, $B \in \mathbb{R}^{k \times n}$; then the standard definition implies

$$(AB)_{ij} = \big(i^{\text{th}} \text{ row of } A\big)^\top \big(j^{\text{th}} \text{ column of } B\big).$$

But matrix multiplication can also be written via outer products as

$$AB = \sum_{i=1}^{k} \big(i^{\text{th}} \text{ column of } A\big)\big(i^{\text{th}} \text{ row of } B\big)^\top.$$

(2) $\Rightarrow$ (3) Hence, $LL^\top = \sum_{i=1}^{n} \big(i^{\text{th}} \text{ column of } L\big)\big(i^{\text{th}} \text{ column of } L\big)^T$. So to mirror the spectral decomposition, we can choose

$$L = \Big[\sqrt{\lambda_1} u_1, \sqrt{\lambda_2} u_2, ..., \sqrt{\lambda_n} u_n\Big],$$

which is valid because eigenvalues are non-negative.

(3) $\Rightarrow$ (4) Here we appeal to the standard interpretation of matrix products. In particular, we have

$$S_{ij} = \big(LL^\top\big)_{ij} = \big(i^{\text{th}} \text{ row of } L\big)^\top \big(j^{\text{th}} \text{ row of } L\big),$$

which means we can simply choose v_i to be the ith row of L.

(4) $\Rightarrow$ (1) Note that $x^\top Sx = \sum_{i,j} x_i x_j S_{ij} = \sum_{i,j} x_i x_j v_i^\top v_j = \left\|\sum_{i=1}^{n} x_i v_i\right\|^2 \geq 0.$ □

Let's try to make sense of (5) by proving its equivalence for 2x2 matrices. Take

$$S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

(5) says $a, c \geq 0$, $ac - b^2 \geq 0$. Recall that λ's are roots of

$$0 = |S - \lambda I| = (a - \lambda)(c - \lambda) - b^2 = \lambda^2 - (a + c)\lambda + (ac - b^2).$$

Note $\lambda_1 \lambda_2 = ac - b^2 \geq 0$. Also $\lambda_1 + \lambda_2 = a + c \geq 0$. Hence $\lambda_1, \lambda_2 \geq 0 \Rightarrow S \succeq 0$.

(1) says $S \succeq 0 \Rightarrow e_1^\top S e_1 = a \geq 0, e_2^\top S e_2 = c \geq 0$. Also $\lambda_1, \lambda_2 \geq 0 \Rightarrow \lambda_1 \lambda_2 = \det(S) = ac - b^2 \geq 0$.

In fact, the last of these works more generally. One can think of isolating a principal submatrix M as choosing a binary matrix P with exactly one non-zero entry per row so that $M = PSP^\top$. Clearly, if $S \succeq 0$ then $M \succeq 0$, and the determinant of M, that is, the product of its eigenvalues, must be non-negative.

The more magical bit is the opposite direction: if the determinant of $PSP^\top$ is non-negative for arbitrary P, that would be enough to get $S \succeq 0$. (Why?) The principal submatrix condition says that checking this for binary P's with one non-zero entry per row is enough.

8. Second-Order Conditions for Local Minimization

We are back to the unconstrained land. Our aim this time is to get both necessary and sufficient conditions for local minima, although not both at the same time.

Proposition 32 x is a local min for $f \Rightarrow \nabla f(x) = 0$ and $\nabla^2 f(x) \succeq 0$.

Proof. We already know $\nabla f(x) = 0$ is necessary. Say $\nabla f(x) = 0$ and $\nabla^2 f(x) \not\succeq 0$. Then there exists u for which $u^T \nabla^2 f(x) u < 0$ and $\|u\| = 1$. Consider $y = x + \alpha u$.

$$f(x + \alpha u) = f(x) + \langle \nabla f(x), u \rangle \alpha + \frac{1}{2}\alpha^2 u^T \nabla^2 f(x) u + r(\alpha, x, u) \quad \text{where} \quad r(\alpha, x, u) = o(\alpha^2)$$

Hence, there exists $\overline{\alpha} > 0$ such that for all $0 < \alpha < \overline{\alpha}$, $|r(\alpha, x, u)| \leq \frac{1}{4} |\alpha^2 u^T \nabla^2 f(x) u|$. Now

$$f(x + \alpha u) \leq f(x) + \frac{1}{4} u^T \nabla^2 f(x) u \alpha^2 < f(x).$$

Yet, $x + \alpha u$ lives in an arbitrarily small neighborhood of x since this holds for all $0 < \alpha < \overline{\alpha}$. Thus, x cannot be a local min. □

Proposition 33 $\nabla f(x) = 0$ and $\nabla^2 f(x) \succ 0 \Rightarrow x$ is a strict local min at x.

Proof. Consider any $y = x + d\varepsilon$ with $\varepsilon \neq 0$ and unit vector d.

$$f(y) = f(x) + \varepsilon \langle \nabla f(x), d \rangle + \frac{\varepsilon^2}{2} d^T \nabla^2 f(x) d + r(x, \varepsilon, d) \quad \text{where} \quad r = o(\varepsilon^2)$$

For any fixed d, there exists $\overline{\varepsilon} > 0$ such that for all $0 < \varepsilon < \overline{\varepsilon}$ we have $|r(x, \varepsilon, d)| \leq \frac{\varepsilon^2}{4} |d^T \nabla^2 f(x) d|$. So $f(y) \geq f(x) + \frac{\varepsilon^2}{4} d^T \nabla^2 f(x) d > f(x)$. The one complication is that this choice of $\overline{\varepsilon}$ depends on d.

To fix this, define $g(d) = \sup\{\overline{\varepsilon} : \forall 0 < \varepsilon < \overline{\varepsilon}, f(x + \varepsilon d) > f(x)\}$, which is continuous. Clearly $g(d) > 0$ for any d; compactness of the set of all unit-norm d's implies that $\min_{\|d\|=1} g(d)$ exists, and hence, is positive. Thus, $f(y) > f(x)$ for all $y \in B\Big(x, \min_{\|d\|=1} g(d)\Big) - \{x\}$. □

8.1. Why not the constrained case?

It is also possible to define second-order conditions (sufficient and necessary) for constrained local minima. However, their practical utility is limited by concerns about computational complexity.

For example, is $x = 0$ a local min for some $f(x)$ on the non-negative orthant $\mathbb{R}_{\geq 0} = \{x : x \geq 0\}$?

To establish necessary conditions, one would hope for:
1. $\langle \nabla f(0), x \rangle \geq 0$ for all $x \in \mathbb{R}_{\geq 0}$ which is the same as $\nabla f(0) \geq 0$.
2. And for all $d \geq 0$ such that $\langle \nabla f(0), d \rangle = 0$, we have $d^\top \nabla^2 f(0) d \geq 0$.

The intuition behind (2) is that for all directions one could move from $x = 0$ without changing the first-order expansion, the second-order term is non-negative.

But, also, checking given a matrix A if for all $x \geq 0$, $x^\top A x \geq 0$ is NP-hard, which is something you would need to do with $\nabla f(0) = 0$. This is called the copositivity problem. Copositive matrices are of course a superset of PSD matrices, but they only attest to non-negativity for non-negative vectors, and checking membership within this set is computationally hard.

9. A Few Topological Definitions

Definition 34 Closure of a set $\mathrm{cl}(A)$ is the smallest closed set containing A.

Here *smallest* is defined in the sense of set inclusion. So two sets may not be comparable, yet the smallest in this sense is well defined.

Definition 35 x is an interior point of A if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$.

Interior of a set $\mathrm{int}(A)$ is the set of all interior points of A.

Definition 36 Boundary of a set $\mathrm{bd}(A) = \mathrm{cl}(A) - \mathrm{int}(A)$.

As an example, consider $\mathbb{Q}$, the set of rational numbers. $\mathrm{cl}(\mathbb{Q}) = \mathbb{R}$, $\mathrm{int}(\mathbb{Q}) = \varnothing$ since any small ball around a rational contains an irrational and $\mathrm{bd}(\mathbb{Q}) = \mathbb{R}$.

As a second example, consider the open ball $A = \{x : \|x\| < 1\}$. Then $\mathrm{cl}(A) = \{x : \|x\| \leq 1\}$ is the closed ball, $\mathrm{int}(A) = A$ and $\mathrm{bd}(A) = \{x : \|x\| = 1\}$ is the sphere.

The annoying bit about these definitions is that the (ambient) space you are in matters. Consider the same set $A = \{x : x_1^2 + x_2^2 < 1, x_3 = 0\}$ in $\mathbb{R}^3$. Now, $\mathrm{cl}(A) = \{x : x_1^2 + x_2^2 \leq 1, x_3 = 0\}$ same as before, but $\mathrm{int}(A) = \varnothing$ since any 3d ball cannot be in the flat set A; $\mathrm{bd}(A) = \mathrm{cl}(A)$. Thus, every point in A is now on the boundary. To avoid such nuisances, we will consider notions that are robust to such changes in description.

Recall that A is an affine space if it can be written as $A = x_0 + V$ for some point x_0 and subspace V.

Definition 37 x is a relative interior point of A if there exists $\varepsilon > 0$, $B(x, \varepsilon) \cap \mathrm{aff}(A) \subseteq A$, where $\mathrm{aff}(A)$ is the smallest affine space containing A.

The relative interior of A, $\mathrm{relint}(A)$, is the set of all relative interior points of A.

Definition 38 The relative boundary of a set A is $\text{rbd}(A) = \text{cl}(A) - \text{relint}(A)$.

Thus for $A = \{x : x_1^2 + x_2^2 < 1\}$, $\text{relint}(A) = A$, $\text{rbd}(A) = \{x : x_1^2 + x_2^2 = 1\}$ regardless of the space in which A is embedded.

10. The Separation Theorem

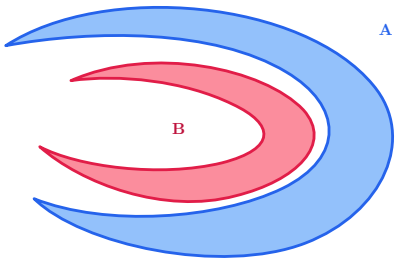
We will now prove one of the most fundamental properties of convex sets.

Definition 39 C and D are separable if there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x \leq b$ for all $x \in C$ and $a^T x \geq b$ for all $x \in D$.

Similarly, C and D are strictly separable if there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x < b$ for all $x \in C$ and $a^T x > b$ for all $x \in D$.

Theorem 40 Non-empty convex sets C and D can be separated if $\text{relint}(C) \cap \text{relint}(D) = \varnothing$.

This theorem in fact holds with "iff" if we strengthen the definition of separability so that at least one of C, D is not contained in the set $\{x : a^\top x = b\}$. Why do we need this additional condition? Note relint of a point is itself. Yet, as per the original definition, a point is separable from itself.



As seen above, convexity of *both* sets is necessary for this result. (Imagine B as an ellipse.)

This is a pretty fundamental result about convex sets. We will state a couple of corollaries that are convenient and then prove a weakened version.

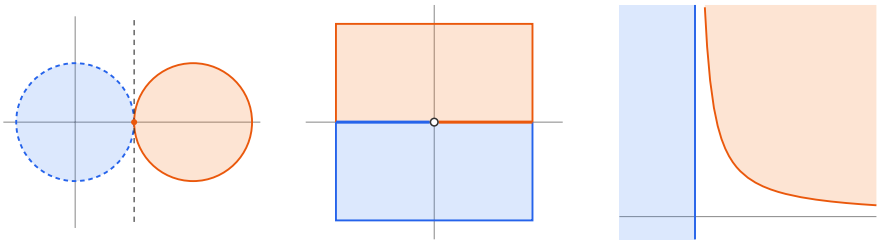
Corollary 40.1 Non-empty closed convex sets C and D such that $C \cap D = \varnothing$ are separable.

Corollary 40.2 Any non-empty convex set is separable from a point on its boundary.

Theorem 41 If C and D are closed convex non-empty sets such that $C \cap D = \varnothing$ and at least one of C, D is bounded, then there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x < b$ for all $x \in C$ and $a^T x > b$ for all $x \in D$.

We will prove this result. This is weaker than the separation theorem in many ways, but also implies strict separation.

Let us examine why each condition is required.



Example. Take C to be the open unit diesk centered at $(0,0)$ and D to be the closed unit disk centered at $(2,0)$. These are convex, bounded, and $C \cap D = \varnothing$. Yet they are not strictly separable.

You might say that maybe being closed is not required if we weaken strict separability to strict inequality only on one side. That does not work either.

Example. In two dimensions, take $C = (\{x : x_2 < 0\} \cup \{x : x_2 = 0, x_1 < 0\}) \cap \{x : \|x\|_\infty \leq 1\}$, $D = (\{x : x_2 > 0\} \cup \{x : x_2 = 0, x_1 > 0\}) \cap \{x : \|x\|_\infty \leq 1\}$, which are convex and bounded, with $C \cap D = \varnothing$. Yet they are not strictly separable (even under the one-sided definition).

Similarly, one of the sets being bounded is necessary.

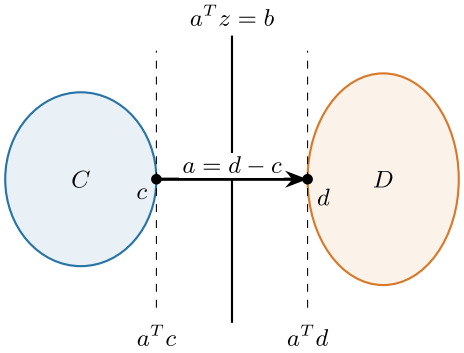
Example. In two dimensions, take $C = \{x : x_1 \leq 0\}$, $D = \{x : x_1 x_2 \geq 1, x_1 \geq 0\}$, which are closed and convex, with $C \cap D = \varnothing$. Yet they are not strictly separable.

Proof. Generically, say C is bounded. Consider $f(x) = \inf_{d \in D} \|x - d\|$ over $x \in C$. Now C is bounded, f is continuous (Why?). Hence, $\min_{x \in C} f(x)$ is achieved at some $c \in C$; let's fix this c.

Now, consider $f(x) = \|x - c\|$ over $x \in D$. Now f is continuous with bounded sub-level sets, so min of $f(x)$ over $x \in D$ is attained. Hence let $(c, d) = \operatorname{argmin}_{c \in C, d \in D} \|c - d\|$.

The idea hereafter is simple. The bisector of the two points is the hyperplane we are looking for. Take $a = d - c \neq 0$ since else $C \cap D \neq \varnothing$ and $b = \frac{\|d\|^2 - \|c\|^2}{2}$. By construction, $a^\top \left(\frac{c+d}{2}\right) = b$, or equivalently, $\frac{c+d}{2}$ lies on the hyperplane.

We claim $a^\top x < b$ for all $x \in C$ and $a^\top x > b$ for all $x \in D$. This might seem obvious, but I encourage you to try to complete the proof before reading further.



Choose any $d' \in D$. Since $f(x) = \frac{1}{2}\|x - c\|^2$ is minimized at d, from Proposition 27, we know that

$$\langle \nabla f(d), d' - d \rangle = \langle d - c, d' - d \rangle \geq 0, \quad \forall d' \in D.$$

Using this, we get

$$\begin{aligned} a^\top d' &= (d-c)^\top d' \geq (d-c)^\top d \\ &= (d-c)^\top \left(\frac{d+c}{2} + \frac{d-c}{2} \right) = b + \frac{\|c-d\|^2}{2} > b. \end{aligned}$$

This works (anti)symmetrically for any $c' \in C$. □

11. Equivalent Conditions for Convexity

Proposition 42 A differentiable function f is convex on a convex set K if and only if, for all $x, y \in K$, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Proof. $\Leftarrow$ Consider any $x, y \in K$, and $\lambda x + (1-\lambda)y \in K$ for some $\lambda \in [0,1]$. Now

$$f(x) \geq f(\lambda x + (1-\lambda)y) + (1-\lambda)\langle \nabla f(\lambda x + (1-\lambda)y), x - y \rangle,$$

$$f(y) \geq f(\lambda x + (1-\lambda)y) + \lambda \langle \nabla f(\lambda x + (1-\lambda)y), y - x \rangle.$$

Add (1)$\times\lambda$ + (2)$\times(1-\lambda)$ to get $\lambda f(x) + (1-\lambda) f(y) \geq f(\lambda x + (1-\lambda)y)$.

$\Rightarrow$ We know $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$, and hence

$$\langle \nabla f(y), x - y \rangle = \lim_{\lambda \to 0^+} \frac{f(y + \lambda(x-y)) - f(y)}{\lambda} \leq f(x) - f(y).$$

□

Proposition 43 A differentiable function f on a convex set K is convex if and only if, for all $x, y \in K$, we have

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0.$$

Proof. $\Rightarrow$ We know

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle,$$

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle.$$

Add these.

$\Leftarrow$ Using the fundamental theorem of calculus, we have

$$
\begin{aligned}
f(y) &= f(x) + \int_0^1 \langle \nabla f(x + \lambda(y-x)), y - x \rangle \, \mathrm{d}\lambda \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \langle \nabla f(x + \lambda(y-x)) - \nabla f(x), y - x \rangle \, \mathrm{d}\lambda \\
&\geq f(x) + \langle \nabla f(x), y - x \rangle.
\end{aligned}
$$

□

Proposition 44 A twice-differentiable function f is convex on a convex set K with non-empty interior if and only if, $\forall x \in \operatorname{int}(K)$, we have

$$\nabla^2 f(x) \succeq 0.$$

Proof. $\Leftarrow$ Note that

$$
\begin{aligned}
f(y) &= f(x) + \int_0^1 \langle \nabla f(x + \lambda(y-x)), y - x \rangle \, \mathrm{d}\lambda \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \int_0^\lambda (y-x)^\top \nabla^2 f(x + \mu(y-x))(y-x) \, \mathrm{d}\mu \, \mathrm{d}\lambda \\
&\geq f(x) + \langle \nabla f(x), y - x \rangle.
\end{aligned}
$$

$\Rightarrow$ Consider any $x \in \operatorname{int} K$; then there exists $\overline{\varepsilon} > 0$ such that $\forall 0 < \varepsilon < \overline{\varepsilon}, x + \varepsilon d \in K$ for any unit vector d. Now, let $y = x + \varepsilon d$.

$$
\begin{aligned}
0 &\leq \frac{\langle \nabla f(y) - \nabla f(x), y - x \rangle}{\varepsilon^2} \\
&= \frac{1}{\varepsilon} \langle \nabla f(y) - \nabla f(x), d \rangle \\
&= \frac{1}{\varepsilon} \int_0^\varepsilon d^\top \nabla^2 f(x + \mu d) d \, \mathrm{d}\mu.
\end{aligned}
$$

Take $\lim_{\varepsilon \to 0^+}$. Since this holds for all d, we are done. (Why?) □

11.1. First-Order Local-to-Global for Convex Functions

Finally, our efforts have paid off. We have the following local condition for convexity that can be cheaply checked and characterizes global optimality.

Corollary 44.1 Let f be a differentiable convex function on K. Then, $\overline{x}$ is a global minimum on K if and only if

$$\langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0, \quad \forall x \in K.$$

This follows by combining Proposition 27 and Proposition 42.

Also, conveniently, for unconstrained optimization, or more generally when $\overline{x} \in \text{int}(K)$, this condition is equivalent to $\nabla f(\overline{x}) = 0$.

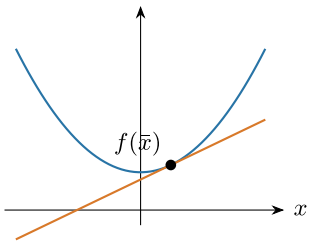
11.2. Subgradients and Their Existence

Convex univariate functions are differentiable everywhere except at countably many points. However, functions of the latter form might not be convex at all, e.g. a step function.

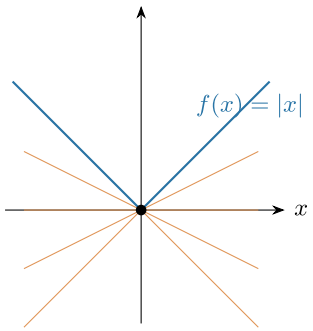
In convex analysis, there is an alternative way to make this concrete with a global definition, that generalizes gradients.

Definition 45 g is a subgradient of f at x on K if for all $y \in K$, $f(y) \geq f(x) + g^{T(y-x)}$.

The set of all subgradients of f at x on K is called the subdifferential $\partial f(x)$.



One supporting slope
$g = f'(\bar{x})$



Many supporting slopes
$\partial f(0) = [-1, 1]$

Theorem 46 Let f be a function on a convex set K. It is true that:
1. If for all $x \in K$ $\partial f(x) \neq \varnothing$, then f is convex, and
2. If f is convex, then, for all $x \in \text{int } K$, $\partial f(x) \neq \varnothing$.

Proof. (1) Same proof as $\Leftarrow$ for Proposition 42.

(2) Take any $x \in \text{int}(K)$. Then $(f(x), x)$ is on the boundary of $\text{epi}(f)$. Thus, $\exists(\alpha, \beta) \neq 0$ such that

$$\alpha t + \beta^\top y \geq \alpha f(x) + \beta^\top x \quad \forall (t, y) \in \text{epi}(f).$$

We claim $\alpha > 0$. If so, taking $t = f(y)$, $f(y) \geq f(x) + \left(-\frac{\beta}{\alpha}\right)^\top (y - x)$, and $\partial f(x) \neq \varnothing$.

To finish up: First, $\alpha \geq 0$ follows from taking $t \to \infty$, which is compatible with the epigraph. Finally, there exists $\varepsilon > 0$ such that $y = x + \varepsilon d \in K$ for all unit vectors d. Then, if $\alpha = 0$, note that $\beta^\top (y - x) = \beta^\top d\varepsilon \geq 0$ for all unit vectors d which can only happen if $\beta = 0$, but both cannot be zero simultaneously. □

Exercise. Prove that:
1. Non-negative sums of convex functions are convex.
2. $f(Ax + b)$ is convex for any convex f.
3. $\max_y f(x, y)$ is convex if for all y, $x \mapsto f(x, y)$ is convex.
4. $g(f(x))$ is convex if f is convex and g is convex nondecreasing.

5. $\min_y f(x, y)$ is convex if f is jointly convex in (x, y).
6. $g(x, y) = yf\left(\frac{x}{y}\right)$ on $\{(x, y) : y > 0\}$ is convex if f is convex on $\mathbb{R}^n$.

47-851: Modern Convex Optimization Lecture 3

Conic Programming and Duality

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12. From Linear Programs to Conic Programs

Let's recall linear programming duality. We'll build up to conic duality in the next two lectures, and this will serve as a good comparison.

$$\textbf{Dual: } \min c^\top x,\ Ax \geq b. \quad \textbf{Primal: } \max b^\top y,\ A^\top y = c,\ y \geq 0.$$

Observation: For any primal feasible y and dual feasible x, $c^\top x \geq b^\top y$. This is called weak duality.

Theorem 47 These conditions are equivalent: (A) dual is feasible & bounded below,
(B) primal is feasible & bounded above,
(C) dual is solvable,
(D) primal is solvable,
(E) both primal & dual are feasible.
Whenever they hold, both primal & dual are equal in value.

$\Rightarrow$ in the dual satisfies $y \geq 0 \equiv y \in K = \mathbb{R}^n_{\geq 0}$.

For all $a, b, c, d \in \mathbb{R}^n$ which satisfies that and $\lambda \in \mathbb{R}_{\geq 0}$, (1) $a \geq a$, (2) $a \geq b$ & $b \geq c \Rightarrow a \geq c$, (3) $a \geq b$ & $b \geq a \Rightarrow a = b$, (4) $a \geq b$ & $c \geq d \Rightarrow a + c \geq b + d$, (5) $a \geq b \Rightarrow \lambda a \geq \lambda b$.

(1) K is a cone. (2) K is pointed, i.e., $a \in K$ & $-a \in K \Rightarrow a = 0$. (3) K is closed under addition. These are related as $a \geq b \equiv a - b \in K$.

Lines underscore equivalence of the stated conditions. Conic programs generalize LPs from $\mathbb{R}^n_{\geq 0}$ to arbitrary, closed convex pointed cones with non-empty interiors. A general conic program involves linear optimization over a feasible set, which is an intersection of an affine space and such a cone.

General conic program: $\max c^\top x$ subject to $Ax = b$, $x \in K$.

13. A Hierarchy of Conic Programs

Examples of admissible cones.
1. $\mathbb{R}^n_{\geq 0}$ gives linear programs.
2. The Lorentz cone $\mathcal{L}^{n+1} = \{(t, x) : t \geq \|x\|_2\}$ gives SOCPs (second-order conic programs).
3. $S^{n\times n}_+ = \{S : S \succeq 0\}$ gives rise to SDPs (semi-definite programs).

Note $\langle A, B\rangle = \mathrm{Tr}(A^\top B)$ below, and $\langle A, A\rangle = \|A\|_F^2 = \sum_{i,j=1}^n A_{ij}^2$.

$$\text{LPs} \subseteq \text{QPs} \subseteq \text{QCQPs} \subseteq \text{SOCPs} \subseteq \text{SDPs}.$$

Convex QPs: $\min x^\top Qx + q^\top x + d$ with $Ax \leq b$.

Convex QCQPs: $\min x^\top Qx + q^\top x + d$ with $x^\top Q_i x + q_i^\top x + d_i \leq 0$ for all $i \in [m]$.

SOCPs: $\min f^\top x$ with $\|A_i x + b_i\|_2 \leq c_i^\top x + d_i$ for all $i \in [m]$.

SDPs: $\min \mathrm{Tr}(CX)$ with $\mathrm{Tr}(A_i X) = b_i$ for all $i \in [m]$, $X \succeq 0$.

Let's talk about SDPs. SDPs are tricky.

Example. SDPs can have infinitely many extreme points.

$$S = \{(x, y) : x^2 + y^2 \leq 1\} = \left\{(x, y) : \begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix} \succeq 0\right\}.$$

Example. For SDPs, feasibility & bounded below $\not\Rightarrow$ solvable. $\min x_1$, $\begin{pmatrix} x_1 & 1 \\ 1 & x_2 \end{pmatrix} \succeq 0$.

Example. Why not impose linear constraints as $Ax = B$? Not all linear constraints can be written this way, e.g., $X_{11} = 0$, which is same as $0 = e_1^\top X e_1 = \mathrm{Tr}(Xe_1e_1^\top)$.

Exercise. Write $\max c^\top x$ subject to $Ax \geq b$ as a SDP.

Exercise. Write a convex generic QCQP as a SOCP. For any $Q = \sum_i \lambda_i u_i u_i^\top \succeq 0$, $Q^{\frac{1}{2}} = \sum_i \sqrt{\lambda_i} u_i u_i^\top \succeq 0$. You might be tempted with

$$x^\top Qx + 2q^\top x + d \leq 0 \equiv \left\|Q^{\frac{1}{2}}x + Q^{-\frac{1}{2}}q\right\|_2 \leq \sqrt{q^\top Qq - d},$$

but this only works if $Q \succ 0$ (strictly). How to handle $Q \succeq 0$?

Also, let's take a second to understand why SOCPs as described can indeed be written as the intersection of an affine space and the Lorentz cone. Clearly,

$$\|Ax + b\|_2 \leq cx + d \equiv (cx + d, Ax + b) \in \mathcal{L}^{n+1}.$$

This is an affine map. Thus, by the fundamental theorem of linear algebra, there exist E, f such that

$$\{(cx + d, Ax + b) : x \in \mathbb{R}^n\} = \{y : Ey = f\}.$$

Recall any (linear) subspace can be expressed as the columnspace of a matrix and as the nullspace of another.

Exercise. Show that any convex optimization problem can be recast as a conic program (for some cone).

Lemma 48 (Schur complement lemma) Let $A \succ 0$. Then $X = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right) \succeq 0$ iff $C - B^\top A^{-1} B \succeq 0$. Schur complement of A in X.

Proof.

$$X = \begin{pmatrix} I & 0 \\ B^\top A^{-1} & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & C - B^\top A^{-1} B \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & I \end{pmatrix}.$$

(Verify!) These are transposes of the other and invertible. Hence, $X \succeq 0$ iff the middle matrix is PSD iff $C - B^\top A^{-1} B \succeq 0$. □

Using this, $\|Ax + b\| \leq cx + d$ iff

$$\begin{pmatrix} (cx+d)I & Ax+b \\ (Ax+b)^\top & cx+b \end{pmatrix} \succeq 0.$$

(Handle the case when $cx + d = 0$ separately.) Thus, SOCPs $\subseteq$ SDPs.

14. Application: Approximation Algorithms for MaxCut

Maximum cut on a graph asks for $\max_{S \subseteq V} |(S \times (V - S)) \cap E|$. $\text{Cut}(S) =$ Edges crossing between S and $V - S$.

For a graph $G = (V, E)$, the Laplacian $L \in \mathbb{R}^{n \times n}$ is such that $L_{ii} = \deg(v_i)$ for all $i \in [n]$ and for all $i \neq j$, $L_{ij} = -1$ if $(v_i, v_j) \in E$. Let $x_S \in \mathbb{R}^n$ such $x_i = 1$ if $i \in S$ else -1.

Clearly,

$$\frac{1}{4} x_S^\top L x_S = \frac{1}{2} \sum_{(i,j) \in E} (1 - x_i x_j) = \text{Cut}(S).$$

Claim 49 $L \succeq 0$.

Proof.

$$x^\top L x = \sum_{i=1}^n \deg(i) x_i^2 - 2 \sum_{(i,j) \in E} x_i x_j = \sum_{(i,j) \in E} (x_i - x_j)^2.$$

□

In fact, consider any binary quadratic program with PSD cost matrix; MaxCut is a special case.

$$\begin{aligned} \text{BQP} &\equiv \max_{x_i^2 = 1 \text{ for all } i \in [n]} x^\top A x \\ &= \max \langle A, x x^\top \rangle \\ &= \max \langle A, X \rangle \end{aligned}$$

with $X_{ii} = 1$ for all $i \in [n]$, $X \succeq 0$, X is rank-1.

Since any rank-1 PSD matrix can be expressed as $vv^\top$.

Define SDP $\equiv \max\langle A, X\rangle$ with $X_{ii} = 1$ for all $i \in [n]$, $X \succeq 0$. Clearly SDP $\geq$ BQP.

Theorem 50 (Nesterov) BQP $\geq \left(\frac{2}{\pi}\right)$ SDP for BQP with $A \succeq 0$.

In fact, the proof of the theorem provides a $\frac{2}{\pi} \approx 0.636$ approximation algorithm for such BQPs. For MaxCut, this SDP can also be used to produce a ≈ 0.878 approximation (which is unimprovable upto UGC). But the above result is more general; we don't use the fact that A is a Laplacian. Also, random cuts, greedy methods, LP relaxations all produce $\frac{1}{2}$ approximation. So, this is not unimpressive.

Proof. Consider any SDP-feasible X. There exist $v_1, ..., v_n \in \mathbb{R}^n$ such that $X_{ij} = v_i^\top v_j$. Since $X_{ii} = 1$, v_i's are unit vectors. Take a random $v \sim \text{Unif}(S^{n-1})$ unit vector. Construct a vector x with $x_i = \text{sign}(\langle v, v_i\rangle)$, where $\text{sign}(z) = \mathbf{1}_{z\geq 0} - \mathbf{1}_{z<0}$. Clearly, x is BQP feasible.

$$\begin{aligned}
\mathbb{E}x_i x_j &= \Pr\big(\text{sign}(v, v_i) = \text{sign}\big(v, v_j\big)\big) - \Pr\big(\text{sign}(v, v_i) \neq \text{sign}\big(v, v_j\big)\big) \\
&= 1 - 2\Pr\big(\text{sign}(v, v_i) \neq \text{sign}\big(v, v_j\big)\big) \\
&= 1 - \frac{4\cos^{-1}\big(\langle v_i, v_j\rangle\big)}{2\pi} \\
&= \frac{2}{\pi}\Big(\frac{\pi}{2} - \cos^{-1}\big(\langle v_i, v_j\rangle\big)\Big) \\
&= \frac{2}{\pi}\sin^{-1}\big(\langle v_i, v_j\rangle\big) = \frac{2}{\pi}\sin^{-1}\big(X_{ij}\big).
\end{aligned}$$

Thus,

$$\text{BQP} \geq \mathbb{E}x^\top Ax = \sum_{i,j} A_{ij}\mathbb{E}x_i x_j = \frac{2}{\pi}\langle A, \sin^{-1} X\rangle,$$

applied elementwise.

Lemma 51 $A \succeq 0$, $B \succeq 0 \Rightarrow \langle A, B\rangle \geq 0$.

Proof. Let $A = \sum_i \lambda_i u_i u_i^\top$.

$$\begin{aligned}
\langle A, B\rangle &= \sum_i \lambda_i \langle u_i u_i^\top, B\rangle \\
&= \sum_i \lambda_i u_i^\top B u_i \geq 0.
\end{aligned}$$

□

Lemma 52 $A \succeq 0$, $B \succeq 0 \Rightarrow A \odot B \succeq 0$. $\odot$ is element-wise product.

Proof. There exist $v_1, ..., v_n$ such for all i, j, $A_{ij} = v_i^\top v_j$. There exist $u_1, ..., u_n$ such for all i, j, $B_{ij} = u_i^\top u_j$. But now

$$\begin{aligned}(A \odot B)_{ij} &= \big(v_i^\top v_j\big)\big(u_i^\top u_j\big) \\ &= \sum_{k,\ell} v_{ik} v_{jk} u_{i\ell} u_{j\ell} \\ &= \Big\langle u_i v_i^\top, u_j v_j^\top \Big\rangle,\end{aligned}$$

vectorizing a matrix. Thus $A \odot B \succeq 0$. □

Corollary 52.1 $A \succeq 0 \Rightarrow A^{\odot n} \succeq 0$ for all $n \in \mathbb{Z}_{\geq 0}$, elementwise powers.

Lemma 53 $\sin^{-1}(X) \succeq X$ for all $X \succeq 0$.

Proof.

$$\begin{aligned}\sin^{-1}(x) &= x + \frac{x^3}{2 \times 3} + (1 \times 3x5)\frac{x^5}{2 \times 4 \times 5} \\ &\quad + (1 \times 3 \times 5)\frac{x^7}{2 \times 4 \times 6 \times 7} + ... \\ &= x + \sum_{k \geq 2} a_k x^k\end{aligned}$$

for some $a_k \geq 0$.

This power series converges to $\sin^{-1}(x)$ uniformly on $[-1, 1]$. Using the last corollary, $\sin^{-1}(X) \succeq X$. □

To complete the proof,

$$\begin{aligned}\text{BQP} &\geq \frac{2}{\pi}\langle A, \sin^{-1}(X)\rangle \\ &= \frac{2}{\pi}(\langle A, \sin^{-1}(X) - X\rangle + \langle A, X\rangle) \\ &\geq \frac{2}{\pi}\ \text{SDP}.\end{aligned}$$

□

15. Application: Stabilizability of Linear Dynamical Systems

Let's do another application, before we move onto SDP duality, and then finally conic duality.

A fundamental question about dynamical system $x_{t+1} = f(x_t)$ is if their long-term behaviors can be neatly categorized. As a concrete question, consider a linear dynamical system $x_{t+1} = Ax_t$. When does such a system end up at 0 (i.e., $x_t \to 0$ as $t \to \infty$) regardless of initialization (x_0)? Such systems are called stable.

Definition 54 A linear dynamical system with transition matrix A is stable if for all $x_0 \in \mathbb{R}^n$, $x_t = A^t x_0 \to 0$ as $t \to \infty$.

Consider a diagonalizable matrix $A = U\Sigma U^{-1}$. Here, U is the matrix of eigenvectors concatenated together. Note that the eigenvectors may not be orthogonal, neither might eigenvalues be real. Any x can be written as $x = \sum_i c_i u_i$. $A^t x = \sum_i \lambda_i^t c_i u_i$. From this we get:

Observation: A is stable iff $\lim_{t\to\infty} A^t = 0$ iff $\rho(A) < 1$.

Here, $\rho(A) = \max_i |\lambda_i|$, modulus. Note, $\rho(\cdot)$ is non-convex.

Example. Consider $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. $\rho(A) = \rho(B) = 0$, $\|A\|_2 = \|B\|_2 = 1$. Yet, $\rho\left(\frac{A+B}{2}\right) = \frac{1}{2}$, $\left\|\frac{A+B}{2}\right\|_2 = \frac{1}{2}$.

Operator/spectral norm: $\|A\|_2 = \max_{\|x\|_2 \le 1} \|Ax\|_2$.

Stability is a property that holds over all trajectories (each with unique starting state). Lyapunov gave a criteria to certify the stability of a system without analyzing all its trajectories. The idea was to find a function $V(x)$ such $V(x) > 0$ for all $x \neq 0$, $V(0) = 0$, $V(f(x)) \le V(x)$ for all x, with strict inequality for $x \neq 0$. Under such conditions, it is clear $V(x_t)$ decreases asymptotically. The nice bit is that he showed such a certificate always exists for "nice-enough" systems. Today, such analysis is called potential-based.

Lemma 55 A is stable iff there exists $P \succeq I$ such $P \succ A^\top P A$.

Proof. $\le$ Take $V(x) = x^\top P x$ where $P \succeq I$, $cP \succeq A^\top P A$ for $0 \le c < 1$. $V(0) = 0$, $V(x) \ge \|x\|^2 > 0$ for all $x \neq 0$ and $V(Ax) = x^\top A^\top P A x \le cV(x)$. Therefore, $V(A^t x) \le c^t V(x) \to 0$ as $t \to \infty$, which also implies $x_t \to 0$ as $t \to \infty$.

$\Rightarrow$ Take $P = \sum_{t\ge 0} (A^t)^\top A^t$. Clearly, $P \succeq I$ as $x_0^\top P x_0 = \sum_{t\ge 0} \|x_t\|^2$. Also

$$A^\top P A = \sum_{t\ge 0} (A^{t+1})^\top A^{t+1} = P - I,$$

hence $P \succ A^\top P A$. The only bit left out is that P as defined has to exist. We use the following fact that follows from Jordan decomposition:

Fact: $\|A^t\| \le O(t^n \rho(A)^t)$.

Hence P is well-defined, as $\rho(A) < 1$. □

Note that $\{P : P \succeq I, cP \succeq A^\top P A\}$ is a SDP feasibility program; and one can do bisection on guess of c in $[0, 1)$.

A more involved question is that of stabilizability. Consider a dynamical system with inputs, $x_{t+1} = Ax_t + Bu_t$. Many systems are not intrinsically stable, but can be stable under correct choice of control inputs. This is what stabilizability captures.

Definition 56 A linear dynamical system (A, B) with inputs is stabilizable if for all $x_0 \in \mathbb{R}^n$, there exist $u_1, u_2, ... \in \mathbb{R}^m$ such $x_t \to 0$ as $t \to \infty$.

The dynamical systems we have considered so far are Markovian; all information at time t is summarized in the state x_t. Using this and linear dynamics, one can gain an alternate characterization.

Observation: (A, B) is stabilizable iff there exists K such $\rho(A + BK) < 1$.

The basic idea is to use inputs of the form $u_t = Kx_t$ regardless of x_t. If so,

$$x_{t+1} = Ax_t + Bu_t = (A + BK)x_t = (A + BK)^t x_0.$$

Thus the dynamical system looks like one without inputs where $A_{\text{eff}} = A + BK$. Since such a K exists with $\rho(A + BK) < 1$ whenever (A, B) is stabilizable, can one find it efficiently?

Note that $\min_K \rho(A + BK)$ is a non-convex program.

Instead, let's try to use our lemma.

$$\begin{aligned}
&(A, B) \text{ is stabilizable} \\
\equiv\ &\text{there exists } K \text{ such } \rho(A + BK) < 1 \\
\equiv\ &\text{there exists } K \text{ such there exists } P \succeq I, \\
&P \succ (A + BK)^\top P(A + BK) \\
\equiv\ &\text{there exists } K \text{ such there exists } P \succeq I, \\
&\begin{pmatrix} P & ((A + BK)P)^\top \\ (A + BK)P & P \end{pmatrix} \succ 0. \quad (1)
\end{aligned}$$

Why? We already know $P \succ 0$, thus by Schur complement lemma, satisfaction of the above constraint is same as

$$P - (A + BK)PP^{-1}P(A + BK)^\top \succ 0.$$

Note this does not match what we had in the last step. But that's fine since what we have as same as $\rho((A + BK)^\top) < 1$, but $(A + BK)^\top$ and $A + BK$ have the same characteristic equation and hence eigenvalues. However we still don't have a SDP at this stage due to non-linearity produced by multiplication of K & P; both are variables.

$$\begin{aligned}
&\equiv \text{there exists } K \in \mathbb{R}^{n \times m},\ P \succeq I, \\
&\begin{pmatrix} P & (AP + BL)^\top \\ AP + BL & P \end{pmatrix} \succ 0. \quad (2)
\end{aligned}$$

At first glance, this might seem like a relaxation, we used $L = KP$ to get a (near) SDP feasibility program. But this is in fact a reformulation since one can go back (i.e. construct a feasible solution for the original program) by setting $K = LP^{-1}$.

For this program, one does once again perform bisection with $\succ$ constraints.

What we were able to accomplish in the last step (an exact reformulation) was a lucky coincidence. As an example, if we want to impose a box constraint on K, we can do so in (1) which was (nominally) non-convex anyways, but now an exact reformulation such as (2) as an SDP is no longer possible. In fact, this problem is NP-hard.

16. Deriving SDP Duals

Claim 57

$$\min_{x\in X}\max_{y\in Y} f(x,y) \geq \max_{y\in Y}\min_{x\in X} f(x,y).$$

If min/max do not exist, interpret these as inf/sup for this section and the next.

Proof. Clearly, for all $x \in X$, $y \in Y$, $f(x,y) \geq \min_{x\in X} f(x,y)$. Also, $\max_{x\in X} g(x) \geq \max_{x\in X} h(x)$ if $g(x) \geq h(x)$ for all $x \in X$. Using this, we get for all $x \in X$,

$$\max_{y\in Y} f(x,y) \geq \max_{y\in Y}\min_{x\in X} f(x,y);$$

hence

$$\min_{x\in X}\max_{y\in Y} f(x,y) \geq \max_{y\in Y}\min_{x\in X} f(x,y).$$

□

Applying this to SDPs:

$$\min_X \operatorname{Tr}(CX) \text{ subject to } \operatorname{Tr}(A_iX) = b_i \text{ for all } i \in [m],\ X \succeq 0 \text{ (primal)}$$

$$= \min_{X\succeq 0}\max_{\lambda\in\mathbb{R}^m} \operatorname{Tr}(CX) + \sum_{i=1}^{m} \lambda_{i(b_i - \operatorname{Tr}(A_iX))}$$

$$= \min_{X\succeq 0}\max_{\lambda\in\mathbb{R}^m} b^\top\lambda + \operatorname{Tr}\left(\left(C - \sum_i \lambda_i A_i\right)X\right).$$

Why? If $b_i = \operatorname{Tr}(A_iX)$, fine. If $b_i > \operatorname{Tr}(A_iX)$, $\lambda_i \to \infty$. If $b_i < \operatorname{Tr}(A_iX)$, $\lambda_i \to -\infty$.

$$\geq \max_{\lambda\in\mathbb{R}^m}\min_{X\succeq 0} b^\top\lambda + \operatorname{Tr}\left(\left(C - \sum_i \lambda_i A_i\right)X\right)$$

$$= \max_{\lambda\in\mathbb{R}^m} \begin{cases} b^\top\lambda \text{ if } C \succeq \sum_i \lambda_i A_i \\ -\infty \text{ otherwise} \end{cases}$$

since $\operatorname{Tr}(YZ) \geq 0$ for all $Y, Z \succeq 0$, and one can choose $X = 0$. Otherwise there exists u such that $u^{\top\left(C - \sum_i \lambda_i A_i\right)}u < 0$ and one can choose $X = \alpha uu^\top$, $\alpha \to \infty$.

$= \max_{\lambda\in\mathbb{R}^m} b^\top\lambda$ subject to $C \succeq \sum_i \lambda_i A_i$ (dual). This is a PSD constraint on an affine map, hence representable as intersection of an affine space and the PSD cone, hence is a SDP.

By our claim, we always used the correct $\geq$ relations. Hence, we have weak duality by construction.

Lemma 58 For any primal feasible X, dual feasible λ, we have $\mathrm{Tr}(CX) \geq b^\top \lambda$.

17. Conic Duality

Consider $\min c^\top x$ subject to $Ax \underset{K}{\geq} b$. Before we get to strong duality, it is worth noting that it's not clear how we even define a dual in the first place. For LPs, our derivation was based on the fact that non-negative combinations of vector inequalities always collapse to valid scalar inequalities. But this is not true for CPs.

Example. $\mathcal{L}^{2+1} = \{(t, x) : t \geq \|x\|_2\}$. Clearly, $(1, 0, -1) \underset{\mathcal{L}^{2+1}}{\geq} (0, 0, 0)$. But

$$(0, 0, 1)^{\top(1,0,-1)} = -1 \not\geq 0 = (0, 0, 1)^{\top(0,0,0)}.$$

You might guess that the multiplier should come from K itself. This is also false.

Example. $K = \{(x, y) : |y| \leq 100x\}$. Then $(1, -100) \underset{K(0,0)}{\geq}$. But $(1, 100) \in K$, yet

$$\langle (1, 100), (1, -100) \rangle = 1 - 100^2 \not\geq 0 = \langle (1, 100), (0, 0) \rangle.$$

The true answer follows from a tautology. For all $y \in K_*$, $a \underset{K}{\geq} b \Rightarrow y^\top a \geq y^\top b$ for all $a, b \in K$.

Definition 59 The dual cone K_* of a set $K \subseteq \mathbb{R}^n$ is $K_* = \{y : \text{for all } x \in K,\ y^\top x \geq 0\}$.

Now, $\min_x c^\top x$ subject to $Ax \underset{K}{\geq} b$ is primal. Take $\max b^\top y$ subject to $A^\top y = c$, $y \in K_*$ to be the dual.

For any feasible pair (x, y),

$$c^\top x = y^\top A x = y^{\top(Ax-b)} + b^\top y \geq b^\top y.$$

Lemma 60 For primal feasible x, dual feasible y, we have $c^\top x \geq b^\top y$.

Lemma 61 Let K be a non-empty set in $\mathbb{R}^n$. Then:
1. K_* is a closed convex cone.
2. $\mathrm{int}(K) \neq \varnothing \Rightarrow K_*$ is pointed.
3. K is a closed convex pointed cone $\Rightarrow \mathrm{int}(K_*) \neq \varnothing$.
4. K is a closed convex cone $\Rightarrow K_{**} = K$.

Proof. (1)

$$K_* = \bigcap_{x \in K} \{y : y^\top x \geq 0\}$$

is intersection of halfspaces (hence convex) and intersection of closed sets (hence closed). Cone is true from definition, because homogeneous inequalities are involved.

(2) Say K_* is not pointed. Then K_* contains an entire line passing through origin, say $\{\lambda a : \lambda \in \mathbb{R}\}$. Then $a^\top x \geq 0$ for all $x \in K$ and $(-a)^\top x \geq 0$ for all $x \in K$. Thus $a^\top x = 0$. But then K is flat, it's entirely contained in the maximal subspace orthogonal to a. Hence, $\text{int}(K) = \varnothing$.

Exercise. Prove part (3) of the above lemma.

(4) First $K \subseteq K_{**}$. Why? Pick any $x \in K$, clearly for all $y \in K_*$, $y^\top x \geq 0$. Thus $x \in K_{**}$. For the other direction, we'll invoke separation. Let's do the contrapositive. Say $x \notin K$. There exists $y \neq 0$ such $y^\top x < y^\top z$ for all $z \in K$. But then $z^\top y$ is lower bounded on K, hence $\min_{z \in K} z^\top y = 0$. Hence $y \in K_*$, and from $y^\top x < 0$, $x \notin K_{**}$. □

Corollary 61.1 K is a closed convex pointed cone with non-empty interior iff K_* is a closed convex pointed cone with non-empty interior.

$$\min c^\top x \text{ subject to } Ax \underset{K}{\geq} b, \text{ where } K \text{ is closed convex pointed cone, } \text{int}(K) \neq \varnothing\text{: primal (P).}$$
$$\max b^\top y \text{ subject to } A^\top y = c,\ y \in K_*\text{: dual (D).}$$

Theorem 62 If P is bounded below and strictly feasible (i.e. $Ax \underset{K}{>} b$), then D is solvable and optimal values match.

By symmetry, if D is bounded above and strictly feasible ($y \underset{K_*}{>} 0$) then P is solvable and optimal values match.

Corollary 62.1 If one of P/D is bounded and strictly feasible, and (x, y) is P/D feasible pair, then (x, y) is P,D optimal iff $b^\top y = c^\top x$ iff $y^{\top(Ax-b)} = 0$.

Proof. Under given conditions, we know optimal values are equal. We also know from weak duality that for any pair of feasible solutions

$$b^\top y \leq D^* \leq P^* \leq c^\top x.$$

Thus attainment of optimal values at (x, y) for (P, D) happens iff $b^\top y = c^\top x$. From the proof of weak duality this is equivalent to $y^{\top(Ax-b)} = 0$, the only inequality being tight. □

Proof of strong duality. Let P^* be the optimal value for the primal problem. We will construct a dual feasible y^* such $b^\top y^* \geq P^*$, via the separation theorem.

Let $V = \{Ax - b : c^\top x \leq P^*\}$. It is non-empty, because any affine/linear map of a halfspace can not be empty. Also $V \cap \text{int}(K) = \varnothing$, otherwise given x such $c^\top x \leq P^*$ and $Ax \underset{K}{>} b$, one can strictly decrease the objective for a close enough feasible point, assuming $c \neq 0$.

Aside note: $c \neq 0$ holds generically. If not, $P^* = 0$. Take $y = 0$, $A^\top y = c$ holds and we have $0 = b^\top y \geq P^*$.

Now, by (actual) separation theorem, there exists $y \neq 0 \in \mathbb{R}^m$ such that

$$\sup_{x \in V} y^\top x \leq \inf_{x \in K} y^\top x.$$

So, $y^\top x$ is bounded below on K. But $\inf_{x \in K} y^\top x$ generically only take values $\{0, -\infty\}$. Thus

$$\sup_{x \in V} y^\top x \leq 0 \leq \inf_{x \in K} y^\top x,$$

and $y \in K_*$ follows from the second inequality. But also the first of these is equivalent to $c^\top x \leq P^* \Rightarrow \left(A^\top y\right)^\top x \leq y^\top b$. This can only happen if $A^\top y = \alpha c$ for some $\alpha \geq 0$. If $\alpha > 0$, then take $y^* = \frac{y}{\alpha} \in K_*$ for which $A^\top y^* = c$ and

$$b^\top y^* \geq \left(A^\top y\right)^\top \frac{x}{\alpha} = c^\top x$$

for all $c^\top x \leq P^*$. We are done. If $\alpha = 0$, $A^\top y = 0$ whereas $b^\top y > 0$ for there exists x, $Ax \underset{K}{\geq} b$. $y \in K_*$ is not null, and hence $y^{\top(Ax-b)} > 0 \Rightarrow c^\top x \leq P^*$ yet $y^\top Ax = 0 > b^\top y$. This is a contradiction because then $c^\top x \leq P^*$ has to empty. □

18. The S-Lemma

Consider a non-convex QCQP $\min_x q_{a(x)} = x^\top Q_a x + q_a^\top x + c_a$ subject to $q_{b(x)} = x^\top Q_b x + q_b^\top x + c_b \geq 0$. Q_a, Q_b are arbitrary, need not be $Q_a \succeq 0$, $Q_b \succeq 0$.

Such programs are efficiently solvable, despite apparent non-convexity.

Theorem 63 Let there exist x such $q_{b(x)} > 0$ strictly. Then for all x, $q_{b(x)} \geq 0 \Rightarrow q_{a(x)} \geq 0$ implies there exists $\lambda \geq 0$ such that for all x, $q_{a(x)} \geq \lambda q_{b(x)}$.

$\leq$ above is evident. Such λ "certifies" the former clause. This has the same spirit as Farkas' lemma or, theorem of alternatives you might have seen before:

$$\min_x q_{a(x)} \text{ subject to } q_{b(x)} \geq 0$$

$$= \max_\gamma \gamma \text{ subject to for all } x,\ q_{b(x)} \geq 0 \Rightarrow q_{a(x)} - \gamma \geq 0$$

$$= \max_{\gamma, \lambda \geq 0} \gamma \text{ subject to for all } x,\ q_{a(x)} - \gamma \geq \lambda q_{b(x)} \text{ (using S-lemma)}$$

$$= \max_{\gamma, \lambda \geq 0} \gamma \text{ subject to for all } x,$$

$$(x, 1)^\top \begin{pmatrix} Q_a - \lambda Q_b & \frac{q_a - \lambda q_b}{2} \\ \frac{(q_a - \lambda q_b)^\top}{2} & c_a - \gamma - \lambda c_b \end{pmatrix} (x, 1) \geq 0$$

$$= \max_{\gamma, \lambda \geq 0} \gamma \text{ subject to}$$

$$\begin{pmatrix} Q_a - \lambda Q_b & \frac{q_a - \lambda q_b}{2} \\ \frac{(q_a - \lambda q_b)^\top}{2} & c_a - \gamma - \lambda c_b \end{pmatrix} \succeq 0.$$

(This is an SDP.)

Proof. This needs a brief justification. $\Rightarrow$ is clear. You might reasonably suspect we are asking for more than is necessary, e.g., $(x,1)^\top M(x,1) \geq 0$ also, in the latter formulation. To complete the argument, say there exists y, $y^\top My < 0$. Then, (1) last coordinate of $y \neq 0$, then by rescaling there exists x such that $(x,1)^\top M(x,1) < 0$. (2) last coordinate of $y = 0$, since $y^\top My$ is continuous, there exists z with non-zero last coordinate with $z^\top Mz < 0$. □

Finally, we'll prove S-lemma. Let's start with something simpler.

Lemma 64 For symmetric A, B, $\mathrm{Tr}(A) \geq 0 > \mathrm{Tr}(B) \Rightarrow$ there exists x such $x^\top Ax \geq 0$ and $x^\top Bx < 0$.

Proof. Two observations: for any $x \in \{-1,1\}^n$, $x^\top Dx = \mathrm{Tr}(D)$ for any diagonal D, since $x_i^2 = 1$. Secondly, for any matrix M,

$$\mathbb{E}_{x\sim \mathrm{Unif}(\{-1,1\}^n)} x^\top Mx = \mathrm{Tr}(M).$$

Now choose $\xi \sim \mathrm{Unif}(\{-1,1\}^n)$, $x = U\xi$, where $B = U\Lambda U^\top$ diagonal.

$$\begin{aligned} x^\top Bx &= \xi^\top U^\top U\Lambda U^\top U\xi \\ &= \mathrm{Tr}(\Lambda) = \mathrm{Tr}(\Lambda U^\top U) = \mathrm{Tr}(B) < 0, \end{aligned}$$

yet

$$\begin{aligned} \mathbb{E}x^\top Ax &= \mathbb{E}\xi^{\top(U^\top AU)}\xi \\ &= \mathrm{Tr}(U^\top AU) = \mathrm{Tr}(AUU^\top) = \mathrm{Tr}(A) \geq 0. \end{aligned}$$

Therefore, there exists x such $x^\top Ax \geq 0$ and $x^\top Bx < 0$. □

Theorem 65 Say there exists $\bar{x}$ such $\bar{x}^\top A\bar{x} > 0$. Then for all x, $x^\top Ax \geq 0 \Rightarrow x^\top Bx \geq 0$ implies there exists $\lambda \geq 0$ such that $B \succeq \lambda A$.

Proof. Consider $\mathrm{OPT} = \min_{x^\top Ax \geq 0} x^\top Bx$. Say for all x, $x^\top Ax \geq 0 \Rightarrow x^\top Bx \geq 0$ and there exists $\bar{x}$, $\bar{x}^\top A\bar{x} > 0$. Then $\mathrm{OPT} = 0$. Consider $\mathrm{OPT}' = \min_{x^\top Ax \geq 0, x^\top x = n} x^\top Bx$. OPT' is also strictly feasible; $\mathrm{OPT}' \geq 0$.

$$\equiv \min_X \mathrm{Tr}(BX) \text{ subject to } \mathrm{Tr}(AX) \geq 0,\ \mathrm{Tr}(X) = n,\ X \succeq 0,\ X \text{ is rank 1.}$$

$$\geq \min_X \mathrm{Tr}(BX) \text{ subject to } \mathrm{Tr}(AX) \geq 0,\ \mathrm{Tr}(X) = n,\ X \succeq 0 \text{ (SDP).}$$

$$\text{SDP dual is } \max_{u,\lambda \geq 0} nu \text{ subject to } B \succeq uI + \lambda A.$$

Exercise. Show that SDP dual is indeed the dual of the SDP.

Let's say we guarantee that SDP ≥ 0 and that strong duality holds. Then, we are done as then there exist $u, \lambda \geq 0$ such $B \succeq uI + \lambda A$ and hence there exists $\lambda \geq 0$, $B \succeq \lambda A$.

Take

$$\bar{X} = \frac{\bar{x}\bar{x}^\top + \alpha I}{\mathrm{Tr}}(\bar{x}\bar{x}^\top + \alpha I)n,$$

for small enough $\alpha > 0$, is strictly feasible for SDP. And for all $\lambda > 0$, there exists u such $B \succ uI + \lambda A$. Thus the SDP dual is also strictly feasible and strong duality holds.

Exercise. In words, describe why adding $x^\top x = n$ as a constraint was essential for this argument.

Now to argue SDP ≥ 0, observe: Compactness ensures that it is solvable; let X^* be the solution. There exists L, $X = LL^\top$. Consider that

$$\mathrm{Tr}(AX^*) = \mathrm{Tr}(L^\top AL) \geq 0,$$

and

$$\mathrm{Tr}(BX^*) = \mathrm{Tr}(L^\top BL) = nu^*.$$

If $u^* < 0$, then by our sign lemma, there exists x such $(Lx)^\top A(Lx) \geq 0$ and $(Lx)^\top B(Lx) < 0$. But this contradicts for all x, $x^\top Ax \geq 0 \Rightarrow x^\top Bx \geq 0$. □

Proof of the original theorem. The idea is to use homogeneous polynomials.

$$\bar{q}_{a(x,t)} = x^\top Q_a x + q_a^\top xt + c_a t^2.$$

$$\bar{q}_{b(x,t)} = x^\top Q_b x + q_b^\top xt + c_b t^2.$$

Given there exists x with $q_{b(x)} > 0$, we have $\bar{q}_{b(x,1)} > 0$.

We'll prove for all (x,t), $\bar{q}_{b(x,t)} \geq 0 \Rightarrow \bar{q}_{a(x,t)} \geq 0$. If so, using our simplified S-lemma, there exists $\lambda \geq 0$, $\bar{q}_{a(x,t)} \geq \lambda \bar{q}_{b(x,t)}$ for all (x,t). Set $t = 1$.

Let's say our sub-claim is untrue. Then, there exists (x,t) such $\bar{q}_{b(x,t)} \geq 0$ but $\bar{q}_{a(x,t)} < 0$. Consider (1) $t \neq 0$. Then substituting $\left(\frac{x}{t}, 1\right)$ results in contradiction. (2) $t = 0$, $\bar{q}_{b(x,t)} > 0$, by continuity there exists (y,s) with $s \neq 0$ with $\bar{q}_{b(y,s)} \geq 0$. Case (1) applies.

Exercise. Argue why the remaining case $t = 0$, $\bar{q}_{b(x,t)} = 0$ results in a contradiction too, completing the proof. □

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Volumetric Algorithms

Lecturer: Karan Singh

47-851: Modern Convex Optimization Lecture 5

First-Order Methods

Lecturer: Karan Singh

47-851: Modern Convex Optimization Lecture 6

Assignment 1

Lecturer: Karan Singh

Q1. Inner Products and Dual Norms (10 points)

Part A. Prove that a norm on a real vector space V is induced by an inner product if and only if it satisfies the parallelogram identity

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \forall x, y \in V.$$

Part B. For a norm $\|\cdot\|$ on $\mathbb{R}^n$, define its dual norm as

$$\|x\|_* = \max_{\|y\| \le 1} y^\top x.$$

1. Calculate the dual norms of $\|\cdot\|_2$ and $\|\cdot\|_\infty$, where $\|x\|_2 = \sqrt{x^\top x}$ and $\|x\|_\infty = \max_i |x_i|$.
2. Show that

$$\partial \|\cdot\|(x) = \{y : \|y\|_* \le 1,\ y^\top x = \|x\|\}.$$

Q2. Local Descent and Checking PSD (10 points)

Part A. Find a function $f : \mathbb{R}^n \to \mathbb{R}$ and a point $x \in \mathbb{R}^n$ such that there is no (non-zero) descent direction at x, and yet x is not a local minimizer.

Part B. Given $x \in \mathbb{R}^n$, form a $(n+1) \times (n+1)$-dimensional matrix

$$X = \begin{bmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ x_1 & x_1 & 0 & \cdots & 0 \\ x_2 & 0 & x_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & 0 & 0 & \cdots & x_n \end{bmatrix}.$$

Show that $X \succeq 0$ if and only if $x_i \ge 0, \forall i \in [n]$ and $\sum_{i=1}^n x_i \le 1$.

Q3. Quasiconvexity and Back (10 points)

Part A. Prove that $f(x) = \log\left(\sum_{i=1}^n a_i e^{x_i}\right)$ where $a_i > 0$ is convex.

Part B. Call a function *kinda-convex* if all its sublevel sets are convex.

1. Give a function that is kinda-convex but not convex.
2. Prove that, for any convex $f : \mathbb{R} \to \mathbb{R}$ and nondecreasing $g : \mathbb{R} \to \mathbb{R}$, $f(g(x))$ is kinda-convex.

3. Can such functions be made convex by stretching the x-axis? More precisely, given a kinda-convex $h : \mathbb{R} \to \mathbb{R}$, does there exist a differentiable, invertible $g : \mathbb{R} \to \mathbb{R}$ such that $h(g(x))$ is convex? Prove your claim.

Challenge (Zero Credits): Try to generalize (A) to $f(X_1, ..., X_n) = \log \operatorname{tr}\left(\sum_{i=1}^{n} a_i e^{X_i}\right)$. Matrix exponential simply exponentiates the eigenvalues in the spectral decomposition.

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Assignment 2

Lecturer: Karan Singh

Q4. Eigenvalues and SDPs (15 points)

Part A. Let $a_i \sim \mathcal{N}(\mu_i, \Sigma_i)$ be multivariate normal random vectors. Recast the following problem as an SDP assuming $1/2 \leq \eta < 1$:

$$\max_x c^T x \quad \text{subject to} \quad \Pr(a_i^T x \leq b_i) \geq \eta \quad (i = 1, ..., m).$$

Part B. Recast the following problem as an SDP:

$$\min_X \frac{\lambda_{\max}(X)}{\lambda_{\min}(X)} \quad \text{subject to} \quad \langle A_i, X \rangle = b_i \quad (i = 1, ..., m), \quad X \succeq 0.$$

Part C. Recast the following problem as an SDP:

$$\min_{X \in \mathbb{S}^n} \sum_{j=1}^{k} \lambda_j(X) \quad \text{subject to} \quad \langle A_i, X \rangle = b_i \quad (i = 1, ..., m),$$

where $\lambda_1(X) \geq ... \geq \lambda_n(X)$ are the eigenvalues.

Q5. Lovász's bound on Shannon capacity (15 points)

Let $\alpha(G)$ be the maximum size of an independent set. Recall

$$\Theta(G) = \lim_{k \to \infty} \sqrt[k]{\alpha(G^k)} = \sup_{k \geq 1} \sqrt[k]{\alpha(G^k)}.$$

Part A. Let J be the all-ones matrix and define

$$\nu(G) = \max_X \big\{ \operatorname{tr}(JX) : \operatorname{tr}(X) = 1,\ X \succeq 0,\ X_{ij} = 0 \text{ if } \{i, j\} \in E \big\}.$$

Show that $\nu(G) \geq \alpha(G)$.

Part B. Show that $\alpha(G_A \boxtimes G_B) \geq \alpha(G_A)\alpha(G_B)$. The strong product has vertex set $V_A \times V_B$. Distinct vertices $(i, k), (j, \ell)$ are adjacent exactly when

$$(\{i, j\} \in E_A \text{ or } i = j) \quad \text{and} \quad (\{k, \ell\} \in E_B \text{ or } k = \ell).$$

Part C. Show that $X, Y \succeq 0$ implies $X \otimes Y \succeq 0$. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$, the Kronecker product $A \otimes B$ is defined as

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mn}B \end{bmatrix}.$$

Part D. Show that $\nu(G_A \boxtimes G_B) \leq \nu(G_A)\nu(G_B)$.

Hint: Use the dual SDP.

Part E. Show that $\nu(G) \geq \Theta(G)$ and use this to show $\nu(C_5) = \Theta(C_5) = \sqrt{5}$.

Hint: Numerically solving the SDP to see if it is close enough to $\sqrt{5}$ is an acceptable proof.

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Assignment 3

Lecturer: Karan Singh

Q6. Grads and Dists (10 points)

Part A: Consider a function $f : \mathbb{R}^d \to \mathbb{R}$ that is μ-strongly convex and L-smooth. Let $x^* \in \arg\min_x f(x)$. Prove for all $x \in \mathbb{R}^d$ that

$$\frac{1}{2L}\|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Part B: The Fenchel conjugate of a function $f : \mathbb{R}^d \to \mathbb{R}$ is defined as $f^*(x) = \max_y(x^\top y - f(y))$. Assuming f^* exists, prove that f is μ-strongly convex if and only if $f^*(x)$ is $\frac{1}{\mu}$-smooth.

Q7. Just Enough Convexity (10 points)

Consider a L-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ that satisfies for all x that

$$f(x) - \min_{x^*} f(x^*) \leq \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Part A: Prove that such a function may not even be convex (much less strongly convex).

Part B: Nevertheless, prove that gradient descent on such a function, with an appropriate step size, produces iterates so that

$$f(x_t) - \min_{x^*} f(x^*) \leq O\Big(e^{-\mu \frac{t}{L}}\Big).$$

Q8. Small Grad Bounds (10 points)

Part A: Fix a convex set $K \subset \mathbb{R}^d$ of diameter D. Recall that online gradient guarantees the following regret bound against a sequence of adversarially chosen convex G-Lipschitz functions f_t.

$$\text{Regret} = \sum_{t=1}^{T} f_t(x_t) - \min_{x^* \in K} \sum_{t=1}^{T} f_t(x^*) \leq O\Big(GD\sqrt{T}\Big),$$

With suitable dynamically chosen step sizes, show that one can guarantee that

$$\text{Regret} \leq O\left(D\sqrt{\sum_{t=1}^{T} \|\nabla f_t(x_t)\|^2}\right).$$

Part B: Assume that all loss functions $f_t : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ are in addition non-negative, well-defined and L-smooth over all of $\mathbb{R}^d$. Our online algorithm must still play points in K and the regret is still defined with respect to the best point in K. Prove that the same algorithm guarantees

$$\text{Regret} \leq O\left(\sqrt{LD^2\left(\min_{x^* \in K} \sum_{t=1}^{T} f_t(x^*)\right)} + LD^2\right).$$

This means whenever there is a point with zero net loss, we get constant regret, independent of T.